**MEMORANDUM**

**TO:** Professor Brodnax
**FROM:** Guo Zhan
**DATE:** October 30, 2026
**RE:** Memo 2: Concept Showcase Plan

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## Concept

In my analysis, artemether-lumefantrine, the largest malaria medicine at USD 0.9 billion, sits among HIV generics in the product network, not with other malaria medicines. That placement raises a question for a buyer: if disease area does not show who could make a medicine, what does?

My showcase will highlight the results of my Insight Report, and the answer it offers is shared suppliers. Louvain community detection (Blondel et al., 2008) found seven communities with a modularity of 0.31, above the 0.22 average from 200 shuffled networks. This structure also helps predict entry. Density (Hidalgo et al., 2007) measures how close a medicine is to what a firm already supplies. Three-year entry rose from 0.15% at zero density to 8.1% above 0.5.

Market shaping starts from unmet need, and having one or two suppliers does not show a problem by itself. Purchase records cannot show unmet need, so I will use WHO's latest invitations for 71 priority medicines as a public signal. For 39, a firm outside the current suppliers is close. For eight, none is, so a new source may need more active support. For 22, the Global Fund reported no recent purchases, and the network cannot say. The audience is a market-shaping lead who has confirmed a need and wants to know where to start.

## Medium

The final product will be an interactive website, The Medicine Supplier Map, built with D3.js from my output tables and submitted as a web link. It opens with a step-by-step game that needs no medical background:

1. The mission. It is 2018, and the reader must get a once-daily HIV pill to more people. Real organizations rarely partner with a company they doubt, so the reader first chooses whether speed, scale or price matters most.

2. Three yearly choices. With names hidden, the reader compares company profiles from my network analysis, spends limited checks on WHO records, then picks a partner. Companies that start first become unavailable.

3. The result. Dragging a timeline reveals real deliveries. The site then scores the plan on all three goals, because no plan wins on everything.

4. The next call. For medicines WHO requests today, the reader sees which companies call for supply talks and which for approval support.

The evidence and a medicine lookup follow the game.

The site will state its limits. The result shows what companies really did, not what a partnership caused. WHO's list signals priority, not a shortage, and predictions rest on only 34 test entries.

I chose this medium because making choices under time pressure makes supplier proximity easy to grasp.

## References


Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment, 2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Hidalgo, C. A., Klinger, B., Barabási, A.-L., & Hausmann, R. (2007). The product space conditions the development of nations. *Science, 317*(5837), 482–487. https://doi.org/10.1126/science.1144581